Bo Tang

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RESEARCH INTERESTS

Decision-focused learning and differentiable optimization; learning to optimize mixed-integer problems; first-order methods and GPU-accelerated optimization.

ACADEMIC APPOINTMENTS

Massachusetts Institute of Technology, Cambridge, MA
Postdoctoral Associate — Jan. 2026 – Current
Advisor: Haihao Lu

EDUCATION BACKGROUND

University of Toronto, Toronto, Canada — GPA: 4.0/4.0
Ph.D. in Mechanical and Industrial Engineering — Sept. 2020 – Dec. 2025
Advisor: Elias B. Khalil
Dissertation: Bridging Machine Learning and Operations Research: From Decision-Focused Learning to Learning to Optimize

Northeastern University, Boston, MA — GPA: 4.0/4.0
M.Sc. in Mathematics — Sept. 2017 – Dec. 2019
Advisor: Mehdi Behroozi

University of International Business and Economics, Beijing, China — GPA: 3.54/4.0
B.Ec. in Actuarial Science — Sept. 2013 – June 2017

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Publications

Bo Tang, Elias Khalil. "PyEPO: A PyTorch-based End-to-End Predict-then-Optimize Library for Linear and Integer Programming." *Mathematical Programming Computation*, 16, 297–335, 2024. [Code, Slides]

Bo Tang, Elias Khalil. "CaVE: A Cone-Aligned Approach for Fast Predict-then-optimize with Binary Linear Programs." *International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR)*, 193–210, 2024 [Code, Slides]

Bo Tang, Elias Khalil. "Multi-Task Predict-then-Optimize." *Learning and Intelligent Optimization Conference (LION)*, 506–522, 2023. [Code]

Ruobing Shen, **Bo Tang**, Leo Liberti, Claudia D'Ambrosio, Stéphane Canu. "Learning Discontinuous Piecewise Affine Fitting Functions using Mixed Integer Programming over Lattice" *Journal of Global Optimization*, 81, 85–

108, 2021. [Code]

Ruobing Shen, **Bo Tang**, Andrea Lodi, Andrea Tramontani, Ismail Ben Ayed. "An ILP Model for Multi-label MRFs with Connectivity Constraints." IEEE *Transactions on Image Processing*, 29, 6909–6917, 2020. [Code]

Preprints and Manuscripts Under Review

Haihao Lu, **Bo Tang**. "H-PDLP: GPU-Accelerated Linear Programming via Generalized PDHG and Homogeneous Self-Dual Embedding." *Preprint, arXiv:2610.17695, 2026*. [Code]

Zhengzhong You, **Bo Tang**, Haoran Liu, Baichuan Mo. "Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation." Under Review at *INFORMS Journal on Computing*.

Bo Tang, Elias Khalil, Ján Drgoňa. "Learning to Optimize for Mixed-Integer Non-linear Programming with Feasibility Guarantees." Major Revision at *INFORMS Journal on Computing*. [Code]

Working Papers

Jonathan Eckstein, Haihao Lu, Gyorgy Matyasfalvi, **Bo Tang**, Chang Yu. "POLARS: A Portable Linear Algebra Runtime for First-Order Optimization." *Working Paper*, 2026.

Bo Tang, Honghui Zheng, Jingyi Zhao, Ján Drgoňa. "NOIR: A MILP and MINLP Solver using Neural Overparameterization for Integer Representation." *Working Paper*, 2026.

RESEARCH SOFTWARE

H-PDLP: GPU-Accelerated LP solver via Homogeneous Self-Dual Embedding [Code]
- Developed and maintained a GPU-accelerated LP solver based on generalized PDHG.
- Created MIPLIB-2017 Infeas, a reproducible benchmark of infeasible and unbounded LP instances.

cuPDLPx: A GPU-Accelerated First-Order LP Solver [Code]
- Developed and maintained solver components and the Python interface.

REINS: A Learning-to-Optimize Framework for Parametric MINLP [Code]
- Developed a learning-based framework for parametric mixed-integer nonlinear optimization.

PyEPO: PyTorch-based End-to-End Predict-then-Optimize Library [Code]
- Developed and maintained an open-source library integrating optimization models with predictive learning.
- Implemented decision-focused learning methods and developed supporting documentation and tutorials.

TALKS, TUTORIALS, AND PRESENTATIONS

Invited Talks and Panels:

INFORMS Annual Meeting – GPU-Accelerating Optimization Across Domains: Invited Panelist	Nov. 2026
Shanghai University of Finance and Economics: Invited Seminar	Aug. 2026
Johns Hopkins University – Center for Systems Science and Engineering: Invited Talk	Apr. 2026

Tutorials:

American Control Conference (ACC) – PIML Workshop: PyEPO Tutorial July 2024

AAAI Conference – LEANOPT Workshop: PyEPO Tutorial Feb. 2024

Cardinal Operations – CORIDM Lecture: Invited Lecture Nov. 2023

Presentations:

INFORMS Annual Meeting: Oral Presentation Nov. 2026

Mixed Integer Programming (MIP) Workshop: Poster Presentation May 2026

INFORMS Annual Meeting: Oral Presentation Oct. 2025

Canadian Operational Research Society (CORS) Annual Conference: Oral Presentations (3) June 2025

Mixed Integer Programming (MIP) Workshop: Poster Presentation June 2025

INFORMS Computing Society (ICS) Conference: Oral Presentation Mar. 2025

INFORMS Annual Meeting: Oral Presentation Oct. 2024

CPAIOR Conference: Oral Presentation May 2024

AAAI Conference – LEANOPT Workshop: Poster Presentation Feb. 2024

INFORMS Annual Meeting: Oral Presentation Oct. 2023

Mixed Integer Programming (MIP) Workshop: Poster Presentation May 2023

Vector Research Symposium: Poster Presentation May 2023

NeurIPS Conference – OPT Workshop: Poster Presentation Dec. 2022

TEACHING EXPERIENCE

MIE365H1: Advanced Operations Research Fall 2020, Fall 2023, Fall 2024

Tutorial Teaching Assistant and Lab Instructor

- Led tutorials and labs, providing instruction on the practical application of advanced algorithms and use of programming tools for solving complex optimization problems. [Notes]

MIE376H1: Mathematical Programming Winter 2024

Tutorial Teaching Assistant and Lab Instructor

- Designed and prepared assignments, exams, and lab sessions to guide students in theory and practice of mathematical modeling.

MIE424H1: Optimization in Machine Learning Winter 2024

Tutorial Teaching Assistant

- Led tutorials focused on deepening students' understanding of machine learning by exploring its foundations through the lens of optimization techniques. [Notes]

INDUSTRY EXPERIENCE

Pacific Northwest National Laboratory, Richland, United States

Data Science Intern, Advanced Computing, Mathematics, and Data Division June 2023 - Sept. 2023

- Contributed to the development of NeuroMANCER, an open-source differentiable programming library.
- Investigated learning-based correction methods for mixed-integer nonlinear optimization.

Hyundai Mobis, Frankfurt Am Main, Germany

Machine Learning Intern, Vision Recognition Algorithm Team Feb. 2019 – Aug. 2019
- Developed image annotation algorithms for segmentation datasets.
- Built a graphical annotation tool to streamline labeling workflows.

Willis Towers Watson, Beijing, China
Actuarial Intern, Investment Risk and Reinsurance Dec. 2016 – May 2017
- Supported actuarial modeling and conducted sensitivity analyses across risk scenarios.

BLOG ENTRIES

When Machine Learning Meets Operations Research: PyEPO Framework and End-to-End Predict-Then-Optimize Algorithms [Chinese Version]

Paper Interpretation and Reproduction: Augmented Lagrangian Method for Block-Structured Integer Programming [English Version | Chinese Version]

A Deep Dive into DeepOPF: A Deep Neural Network Approach for Security-Constrained DC Optimal Power Flow Problems [Chinese Version]

Decision Trees and Mathematical Programming: Three Mixed-Integer Linear Programming Models for Optimal Classification Trees and Their Implementation [Chinese Version]

ACADEMIC SERVICE

Journal Reviewer: INFORMS Journal on Computing, IISE Transactions, Journal of Heuristics.

Conference Reviewer: ICLR 2025–2026; ICML 2024–2025; NeurIPS 2023, 2025.

Program Committee: ScaleOPT: GPU-Accelerated and Scalable Optimization, NeurIPS Workshop, 2025.

Session Chair: Optimization and Robust Linear Models, INFORMS Annual Meeting, 2024.

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